

Mechanisms underlying skeletal muscle insulin resistance induced by fatty acids: importance of the mitochondrial function

Insulin resistance condition is associated to the development of several syndromes, such as obesity, type 2 diabetes mellitus and metabolic syndrome. Although the factors linking insulin resistance to these syndromes are not precisely defined yet, evidence suggests that the elevated plasma free fatty acid (FFA) level plays an important role in the development of skeletal muscle insulin resistance. Accordingly, in vivo and in vitro exposure of skeletal muscle and myocytes to physiological concentrations of saturated fatty acids is associated with insulin resistance condition. Several mechanisms have been postulated to account for fatty acids-induced muscle insulin resistance, including Randle cycle, oxidative stress, inflammation and mitochondrial dysfunction. Here we reviewed experimental evidence supporting the involvement of each of these propositions in the development of skeletal muscle insulin resistance induced by saturated fatty acids and propose an integrative model placing mitochondrial dysfunction as an important and common factor to the other mechanisms.